

A Quality Evaluation Framework for AI-Generated Mathematical Reasoning in STEM Training Datasets

Macksunday Ayoga

Department of Financial Mathematics, University of Nairobi

macksundayonyonka@gmail.com

May 2026

Abstract

The quality of AI mathematical reasoning — its logical validity, notational precision, methodological diversity, and pedagogical clarity — is central to the reliability and educational value of AI systems in STEM contexts. Yet systematic, reproducible frameworks for evaluating such reasoning remain underdeveloped. This paper proposes the five-dimensional Mathematical Reasoning Quality (MRQ) framework for assessing AI-generated solutions across: Logical Validity, Solution Pathway Diversity, Notational Rigor, Error Transparency, and Pedagogical Clarity. Drawing on practical experience training and evaluating large language models on advanced mathematics tasks, we define quantitative rubrics for each dimension and demonstrate their application through annotated case studies spanning differential equations, linear algebra, combinatorial proofs, and real analysis.

Keywords: *AI evaluation · mathematical reasoning · STEM training data · large language models · proof verification · LaTeX · dataset quality · human-AI alignment*

1. Introduction

The emergence of large language models capable of solving advanced mathematics represents a significant milestone in AI. However, benchmark performance alone is insufficient. A model producing the correct final answer through flawed intermediate steps is not exhibiting rigorous mathematical reasoning — it is producing output that resembles a correct solution without embodying correct mathematical thought.

This distinction matters for two reasons. First, reliability: flawed intermediate reasoning increases the probability of compounding errors on novel problems where the model cannot rely on pattern-matching. Second, pedagogy: AI systems used in educational contexts must model rigorous reasoning that students can learn from and trust.

This paper proposes the Mathematical Reasoning Quality (MRQ) Framework — a five-dimensional rubric with a maximum composite score of 20 — and demonstrates its application across four

problem domains.

2. Background and Related Work

2.1 Mathematical AI Benchmarks

Leading benchmarks include MATH (Hendrycks et al., 2021), GSM8K (Cobbe et al., 2021), MathBench (Liu et al., 2024), and FrontierMath (Glazer et al., 2024). These measure solution accuracy but do not evaluate reasoning quality — the logical structure, methodological richness, or notational precision of generated solutions.

2.2 Rubric-Based Assessment in Mathematics Education

Educational rubrics for mathematical work (NCTM Standards for Mathematical Practice) assess strategic competence, reasoning and proof, procedural fluency, and conceptual understanding (Kilpatrick et al., 2001) — dimensions that map naturally to the MRQ framework.

3. The MRQ Framework

The MRQ Framework evaluates AI-generated solutions across five dimensions, each scored 0–4, yielding a maximum composite score of 20.

Dimension	Code	What It Measures
Logical Validity	LV	Each step follows necessarily from prior steps or axioms; no unjustified leaps
Solution Pathway Diversity	SPD	Awareness of multiple valid approaches; appropriateness of chosen method
Notational Rigor	NR	LaTeX precision, quantifier use, consistent variable definitions
Error Transparency	ET	Domain restrictions, convergence conditions, degenerate cases identified
Pedagogical Clarity	PC	Logical sequence, explained transformations, interpreted results

3.1 Dimension 1: Logical Validity (LV)

Score	Criterion
4	Every inference explicitly justified; all conditions checked; no logical gaps
3	Minor gap that does not affect the conclusion
2	One unjustified step that could cause error in a variant problem

Score	Criterion
1	Multiple unjustified steps; partially correct but structurally flawed
0	Logically invalid; conclusion does not follow from premises

Example (induction proof of sum formula): A LV=4 response provides: (i) Base case: verified for $n=1$; (ii) Inductive hypothesis: assume formula holds for m ; (iii) Inductive step: algebraically derive for $m+1$; (iv) Conclusion: by the principle of mathematical induction, the formula holds for all n in N .

3.2 Dimension 2: Solution Pathway Diversity (SPD)

A high-SPD response for solving $dy/dx - y = \exp(x)$ would present: (1) Primary method via integrating factor yielding $y = x \cdot \exp(x) + C \cdot \exp(x)$; (2) Alternative via variation of parameters; (3) Alternative via Laplace transform; plus a note explaining why the integrating factor is most efficient for first-order linear ODEs.

3.3 Dimension 3: Notational Rigor (NR)

Key standards assessed: proper variable definition before use ("Let $\epsilon > 0$ be arbitrary"); correct eigenvalue equation $\det(A - \lambda I) = 0$; proper integral notation with bounds and differential; matrix equation clarity distinguishing scalars from vectors.

3.4 Dimension 4: Error Transparency (ET)

For the system $Ax = b$, a response stating only $x = A^{-1} \cdot b$ without verifying $\det(A) \neq 0$ scores ET = 1. A complete response adds: "This solution holds iff A is nonsingular. If $\det(A) = 0$, the system either has no solution or infinitely many solutions, determined by the rank-nullity theorem."

3.5 Dimension 5: Pedagogical Clarity (PC)

A PC=4 response is logically sequenced, explains key transformations rather than merely stating them, and interprets results rather than just computing them. The distinction between scratchwork and formal proof in epsilon-delta arguments is a key indicator of mathematical maturity.

4. Application: Case Studies

4.1 Optimization — Lagrange Multipliers

Problem: Minimize $f(x,y) = x^2 + y^2$ subject to $x + y = 1$. Evaluation of a representative AI response:

Dimension	Score	Rationale
Logical Validity	4	All KKT conditions applied; second-order check included
Solution Pathway Diversity	2	No alternative method (substitution, geometric) presented

Dimension	Score	Rationale
Notational Rigor	4	Lagrangian properly defined; LaTeX throughout
Error Transparency	3	Constraint qualification not explicitly verified
Pedagogical Clarity	4	Geometric interpretation adds educational value
Total MRQ	17/20	Strong response; minor SPD gap

4.2 Linear Algebra — Eigenvalue Decomposition

For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, a complete response derives the characteristic equation $\det(A - \lambda I) = (2 - \lambda)^2 - 1 = 0$, yielding eigenvalues 1 and 3, computes orthonormal eigenvectors, notes that symmetry of A guarantees real eigenvalues (Spectral Theorem), and discusses the diagonalisation $A = PDP^{-1}$ and its use for computing matrix powers.

4.3 Real Analysis — Epsilon-Delta

Proving $\lim_{x \rightarrow 2} (3x-1) = 5$ requires showing: for all $\epsilon > 0$, there exists $\delta > 0$ such that $|x-2| < \delta$ implies $|(3x-1)-5| < \epsilon$. A LV=4 response derives the choice $\delta = \epsilon/3$ in a scratchwork phase before deploying it in the formal proof — not pulling δ from nowhere.

5. Implications for Dataset Construction

5.1 Prompt-Response Pair Design

Effective mathematical training data requires prompt-response pairs where: the prompt specifies the target reasoning depth; the response demonstrates the full MRQ profile; and contrastive examples (weak vs strong) enable preference learning algorithms (RLHF, DPO) to reward reasoning quality.

5.2 Calibration

For MRQ to function reliably, evaluators must be calibrated against anchor examples. We recommend a set of 25 gold-standard scored solutions (5 per MRQ band) per domain, with inter-rater reliability target of Cohen's kappa ≥ 0.70 across all dimensions before sign-off.

5.3 Automating Partial Scoring

Dimension 3 (Notational Rigor) can be partially automated via LaTeX syntax validators. Dimension 1 (Logical Validity) may be addressable via formal verification on structured problems. This pre-screening layer enables scalable dataset construction for large problem volumes.

6. Conclusion

The MRQ Framework formalises the distinction between a mathematically correct solution and a mathematically excellent one. Training AI systems on high-MRQ datasets produces models better positioned to reason on novel problems, serve as reliable educational tools, and support human mathematical work in research and professional contexts.

Future work will: (1) automate partial MRQ scoring using formal verification; (2) measure the correlation between training data MRQ distribution and downstream benchmark performance; and (3) extend the framework to proof-heavy domains including abstract algebra and combinatorics.

References

- Cobbe, K. et al. (2021). Training verifiers to solve math word problems. *arXiv:2110.14168*.
- Glazer, E. et al. (2024). FrontierMath: A benchmark for evaluating advanced mathematical reasoning in AI. *arXiv:2411.04872*.
- Hendrycks, D. et al. (2021). Measuring mathematical problem solving with the MATH dataset. *arXiv:2103.03874*.
- Kilpatrick, J., Swafford, J., & Findell, B. (Eds.). (2001). *Adding It Up*. National Academy Press.
- Lewkowycz, A. et al. (2022). Solving quantitative reasoning problems with language models. *NeurIPS 2022*.
- Liu, Y. et al. (2024). MathBench: Evaluating LLMs with a hierarchical mathematics benchmark. *arXiv:2405.12209*.

© 2026 Macksunday Ayoga. Released under the MIT License.